INFO20003 Database Systems

Lecturer: Egemen Tanin

Lecture 15
Normalization

Week 8

What's wrong with the *organization* of data in this table?

Student ID#	Student Name	Campus Address	Degree	Phone	Subject ID	Subject Title	Lecturer Name	Lecturer Office	Lecturer Phone	Sem.	Grade
A121	Joy Egbert	166 Grattan Street	B.Com.	555-7771	ACC101	Accounting	Davern	T240C	8344-1846	1-11	H1
A121	Joy Egbert	166 Grattan Street	B.Com.	555-7771	ECO101	Economics	Smyth	T240F	8344-1868	1-11	H2B
A121	Joy Egbert	166 Grattan Street	B.Com.	555-7771	ECO104	Quant. M.	Collier	T240D	8344-5716	1-11	H2B
A121	Joy Egbert	166 Grattan Street	B.Com.	555-7771	FIN101	Finance.	James	T240D	8344-5275	1-11	H2A
A121	Joy Egbert	166 Grattan Street	B.Com.	555-7771	ACC103	Processes	Wise	T240E	8344-5309	1-11	H3
A123	Larry Mueller	302 Royal Parade	B.Com.	555-1235	ACC101	Accounting	Davern	T240C	8344-1846	1-11	H1
A123	Larry Mueller	302 Royal Parade	B.Com.	555-1235	ECO101	Economics	Smyth	T240F	8344-1868	1-11	H2B
A123	Larry Mueller	302 Royal Parade	B.Com.	555-1235	ECO104	Quant. M.	Collier	T240D	8344-5716	1-11	H2A
A123	Larry Mueller	302 Royal Parade	B.Com.	555-1235	FIN101	Finance.	James	T240D	8344-5275	1-11	H3
A124	Mike Guon	224 Swanston St.	B.Eco.	555-2214	ACC101	Accounting	Davern	T240C	8344-1846	1-11	H2A
A124	Mike Guon	224 Swanston St.	B.Eco.	555-2214	ECO101	Economics	Smyth	T240F	8344-1868	1-11	H2A
A124	Mike Guon	224 Swanston St.	B.Eco.	555-2214	ECO104	Quant. M.	Collier	T240D	8344-5716	1-11	H2B
A124	Mike Guon	224 Swanston St.	B.Eco.	555-2214	ACC103	Processes	Wise	T240E	8344-5309	1-11	H2B
A126	Jackie Judson	85 Barry Street	B.Eco.	555-1245	ACC101	Accounting	Davern	T240C	8344-1846	1-11	H1
A126	Jackie Judson	85 Barry Street	B.Eco.	555-1245	ECO101	Economics	Smyth	T240F	8344-1868	1-11	H2B
A126	Jackie Judson	85 Barry Street	B.Eco.	555-1245	ECO104	Quant. M.	Collier	T240D	8344-5716	1-11	H2B
A126	Jackie Judson	85 Barry Street	B.Eco.	555-1245	ACC103	Processes	Wise	T240E	8344-5309	1-11	H2A
...	...	...	...	...	...	...	...	...	...	...	...

Reading: Chapter 19 from Ramakrishnan and Gehrke

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Anomalies in Denormalized Data

- Consider the following denormalized table (relation) :

Student-ID	Course-ID	Fee
130	C200	75
200	C300	100
250	C200	75
425	C400	150
500	C300	100
575	C500	50
…	…	…

- ***Insertion Anomaly***: A new course cannot be added until at least one student has enrolled (which comes first student or course?)
- ***Deletion Anomaly***: If student 425 withdraws, we lose all record of course C400 and its fee!
- ***Update Anomaly***: If the fee for course C200 changes, we have to change it in multiple records (rows), else the data will be inconsistent.

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Normalisation

- A technique used to remove undesired redundancy from databases
- Break one large table into several smaller tables
- A relation is normalized if all determinants are candidate keys

How do we normalise?

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Invoice example

Bill To

John
Synex Inc
128 AA Juanita Ave
Glendora
CA 91740 US

Ship To

John
Synex Inc
128 AA Juanita Ave
Glendora
CA 91740 US

Date	14-Aug-2009	Order No		Sales Person	Charles Wooten
Shipping Date	13-Aug-2009	Shipping Terms		Terms	COD

ID	SKU / Description	Unit Price (USD)	Qty	Amount (USD)
PS.V860.005	AMD Athlon X2DC-7450, 2.4GHz/1GB/160GB/SMP-DVD/VB	580.00	6.00	3,480.00
PS.V880.037	PDC-E5300 - 2.6GHz/1GB/320GB/SMP-DVD/FDD/VB	645.00	4.00	2,580.00
LC.V890.002	LG 18.5" WLCD	230.00	10.00	2,300.00
HP.Q754.071	HP LaserJet 5200	1,103.00	1.00	1,103.00

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Invoice example – Spreadsheet Format

Invoice Number	Date	Customer Name	Customer Address	Sales Person	Terms	Product ID	Product Name	Unit Price	Quantity	Amount	Sub Total
INV0012	14-Aug-09	John / Synex	128 Juanita Ave...	Charles Wooten	COD	PSV880.006	AMD Athlon X2DC	580	6	3480	9463
						PSV880.037	PDC E5300	645	4	2580	
						LC.V890.002	LG 8.5” LCD	230	10	2300	
						HPQ754.071	HP LaserJet 5200	1103	1	1103	
INV0013	15-Aug-09	Mary / ThisCo	123 Smith Street...	Charles Wooten	COD	HP Q754.071	HP LaserJet 5200	1103	2	2206	3356
						LCV890.002	LG 8.5” LCD	230	5	1150	

This is not relational model

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Invoice example – Spreadsheet Format

Invoice Number	Date	Customer Name	Customer Address	Sales Person	Terms	Sub Total	Discount	Sales Tax	Shipping
INV0012	14-Aug-09	John / Synex	128 Juanita Ave...	Charles Wooten	COD	9463	0	780.70	0
INV0013	15-Aug-09	Mary / ThisCo	123 Smith Street...	Charles Wooten	COD	3356	0	100	0

Product ID	Product Name	Unit Price	Quantity	Amount
PSV880.006	AMD Athlon X2DC	580	6	3480
PSV880.037	PDC E5300	645	4	2580
LC.V890.002	LG 8.5” LCD	230	10	2300
HPQ754.071	HP LaserJet 5200	1103	1	1103
HPQ754.071	HP LaserJet 5200	1103	2	2206
LCV890.002	LG 8.5” LCD	230	5	1150

Break into two
But…
How do we connect?

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Invoice example – Spreadsheet Format

Invoice Number	Date	Customer Name	Customer Address	Sales Person	Terms	Sub Total	Discount	Sales Tax	Shipping
INV0012	14-Aug-09	John / Synex	128 Juanita Ave...	Charles Wooten	COD	9463	0	780.70	0
INV0013	15-Aug-09	Mary / ThisCo	123 Smith Street...	Charles Wooten	COD	3356	0	100	0

Product ID	Product Name	Unit Price	Quantity	Amount	*Invoice Number*
PSV880.006	AMD Athlon X2DC	580	6	3480	INV0012
PSV880.037	PDC E5300	645	4	2580	INV0012
LC.V890.002	LG 8.5” LCD	230	10	2300	INV0012
HPQ754.071	HP LaserJet 5200	1103	1	1103	INV0012
HPQ754.071	HP LaserJet 5200	1103	2	2206	INV0013
LCV890.002	LG 8.5” LCD	230	5	1150	INV0013

Add FK

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Invoice example – Spreadsheet Format

Invoice Number	Date	Customer Name	Customer Address	Sales Person	Terms	Sub Total	Discount	Sales Tax	Shipping
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INV0013	15-Aug-09	Mary / ThisCo	123 Smith Street...	Charles Wooten	COD	3356	0	100	0

This is about product

This is about Order (invoice)

Product ID	Product Name	Unit Price	Quantity	Amount	*Invoice Number*
PSV880.006	AMD Athlon X2DC	580	6	3480	INV0012
PSV880.037	PDC E5300	645	4	2580	INV0012
LC.V890.002	LG 8.5” LCD	230	10	2300	INV0012
HPQ754.071	HP LaserJet 5200	1103	1	1103	INV0012
HPQ754.071	HP LaserJet 5200	1103	2	2206	INV0013
LCV890.002	LG 8.5” LCD	230	5	1150	INV0013

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Invoice example – Spreadsheet Format

Invoice Number	Date	Customer Name	Customer Address	Sales Person	Terms	Sub Total	Discount	Sales Tax	Shipping
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INV0013	15-Aug-09	Mary / ThisCo	123 Smith Street...	Charles Wooten	COD	3356	0	100	0

Product ID	**Quantity**	**Amount**	***Invoice Number***
PSV880.006	6	3480	INV0012
PSV880.037	4	2580	INV0012
LC.V890.002	10	2300	INV0012
HPQ754.071	1	1103	INV0012
HPQ754.071	2	2206	INV0013
LCV890.00	5	1150	INV0013

Break into two

Product ID	**Product Name**	**Unit Price**
PSV880.006	AMD Athlon X2DC	580
PSV880.037	PDC E5300	645
LC.V890.002	LG 8.5” LCD	230
HPQ754.071	HP LaserJet 5200	1103

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INV0013	15-Aug-09	Mary / ThisCo	123 Smith Street...	Charles Wooten	COD	3356	0	100	0

Product ID	Quantity	Amount	*Invoice Number*
PSV880.006	6	3480	INV0012
PSV880.037	4	2580	INV0012
LC.V890.002	10	2300	INV0012
HPQ754.071	1	1103	INV0012
HPQ754.071	2	2206	INV0013
LCV890.00	5	1150	INV0013

What about amount?

Product ID	Product Name	Unit Price
PSV880.006	AMD Athlon X2DC	580
PSV880.037	PDC E5300	645
LC.V890.002	LG 8.5” LCD	230
HPQ754.071	HP LaserJet 5200	1103

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Invoice example – Spreadsheet Format

Invoice Number	Date	Customer Name	Customer Address	Sales Person	Terms	Sub Total	Discount	Sales Tax	Shipping
INV0012	14-Aug-09	John / Synex	128 Juanita Ave...	Charles Wooten	COD	9463	0	780.70	0
INV0013	15-Aug-09	Mary / ThisCo	123 Smith Street...	Charles Wooten	COD	3356	0	100	0

What about sales person?

Product ID	Quantity	*Invoice Number*
PSV880.006	6	INV0012
PSV880.037	4	INV0012
LC.V890.002	10	INV0012
HPQ754.071	1	INV0012
HPQ754.071	2	INV0013
LCV890.00	5	INV0013

Could be derived

Product ID	Product Name	Unit Price
PSV880.006	AMD Athlon X2DC	580
PSV880.037	PDC E5300	645
LC.V890.002	LG 8.5” LCD	230
HPQ754.071	HP LaserJet 5200	1103

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Invoice example – Spreadsheet Format

Invoice Number	Date	Customer Name	Customer Address	*Sales Person ID*	Terms	Sub Total	Discount	Sales Tax	Shipping
INV0012	14-Aug-09	John / Synex	128 Juanita Ave...	1	COD	9463	0	780.70	0
INV0013	15-Aug-09	Mary / ThisCo	123 Smith Street...	1	COD	3356	0	100	0

What about customer?

Product ID	Quantity	*Invoice Number*
PSV880.006	6	INV0012
PSV880.037	4	INV0012
LC.V890.002	10	INV0012
HPQ754.071	1	INV0012
HPQ754.071	2	INV0013
LCV890.00	5	INV0013

Sales Person ID	Sales Person
1	Charles Wooten

Product ID	Product Name	Unit Price
PSV880.006	AMD Athlon X2DC	580
PSV880.037	PDC E5300	645
LC.V890.002	LG 8.5” LCD	230
HPQ754.071	HP LaserJet 5200	1103

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Invoice example – Spreadsheet Format

Invoice Number	Date	Customer ID	*Sales Person ID*	Terms	Sub Total	Discount	Sales Tax	Shipping
INV0012	14-Aug-09	1	1	COD	9463	0	780.70	0
INV0013	15-Aug-09	2	1	COD	3356	0	100	0

Product ID	Quantity	*Invoice Number*
PSV880.006	6	INV0012
PSV880.037	4	INV0012
LC.V890.002	10	INV0012
HPQ754.071	1	INV0012
HPQ754.071	2	INV0013
LCV890.00	5	INV0013

Sales Person ID	Sales Person
1	Charles Wooten

Customer ID	Customer Name	Customer Address
1	John / Synex	128 Juanita Ave...
2	Mary / ThisCo	123 Smith Street...

Product ID	Product Name	Unit Price
PSV880.006	AMD Athlon X2DC	580
PSV880.037	PDC E5300	645
LC.V890.002	LG 8.5” LCD	230
HPQ754.071	HP LaserJet 5200	1103

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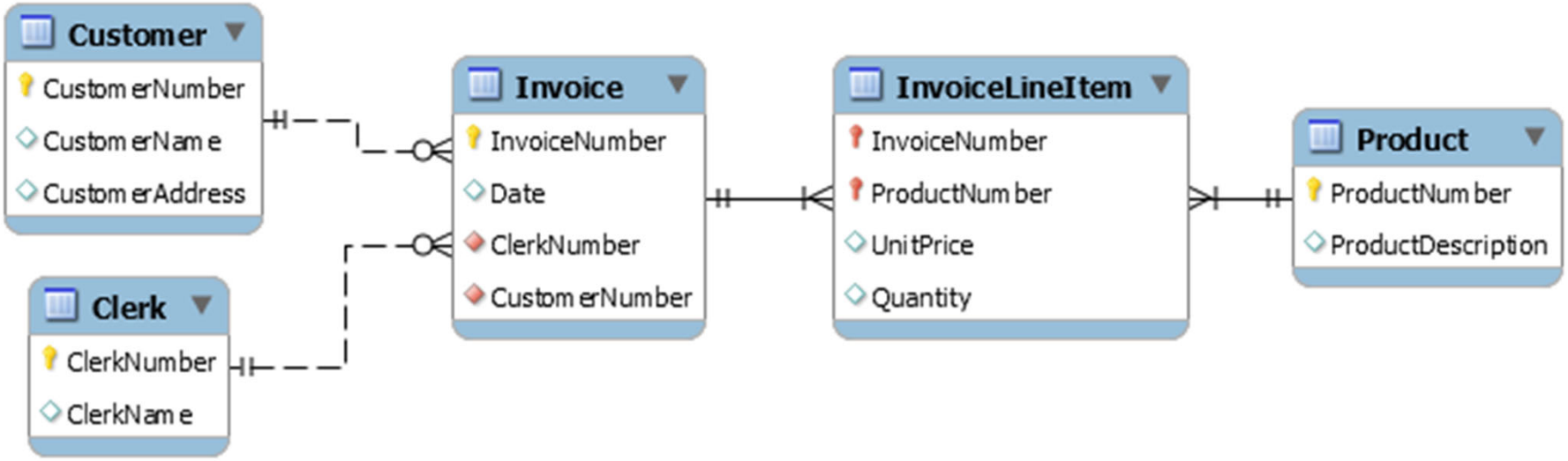
Normalized Relations and ER Diagram

- We can name the relations now
  - Customer (CustomerNumber, CustomerName, CustomerAddress)
  - Clerk (ClerkNumber, ClerkName)
  - Product (ProductNumber, ProductDescription)
  - Invoice (InvoiceNumber, Date, *CustomerNumber*, *ClerkNumber*)
  - InvoiceLineItem (*InvoiceNumber*, *ProductNumber*, UniPrice, Quantity)



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More Formal Concepts: Functional Dependency

- A functional dependency concerns values of attributes in a relation
- A set of attributes X **determines** another set of attributes Y if each value of X is associated with only one value of Y
  - Written X → Y
  - X determines Y (If I know X then I also know Y)

- Emp# → Emp-name
- Emp# → Salary

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Functional Dependency: Definitions

- Determinants $(X, Y \rightarrow Z)$ $A(\underline{X}, \underline{Y}, Z, D)$
  - the attribute(s) on the left hand side of the arrow
- Key and Non-Key attributes
  - each attribute is either part of the primary key or it is not
- Partial functional dependency $(Y \rightarrow Z)$
  - a functional dependency of one or more non-key attributes upon part (but not all) of the primary key
- Transitive dependency $(Z \rightarrow D)$
  - a functional dependency between 2 (or more) non-key attributes

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